

Exercise 1 Let $\mathcal{X} \subset \mathbb{R}^n$ be a polytope with vertices v_i , $i = 1, 2, \dots, m$. Show that $\mathcal{X}$ is an invariant set for the system $x(k+1) = Ax(k)$ if and only if it holds for all $i = 1, 2, \dots, m$ that $Av_i \in \mathcal{X}$.

Hint 1: For a polytope $\mathcal{X} \subset \mathbb{R}^n$ every $x \in \mathcal{X}$ can be written as a convex combination of the vertices $x = \sum_{i=1}^m \lambda_i v_i$, $\sum_{i=1}^m \lambda_i = 1$, $\lambda_i \geq 0$.

Hint 2: Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a convex set and consider vectors $y^1, y^2, \dots, y^k \in \mathcal{X}$. It holds that the convex hull $\text{co}(y^1, y^2, \dots, y^k) \subseteq \mathcal{X}$.

Exercise 2 Consider the MPC problem

$$\begin{aligned} \min_U \quad & px_N^2 + \sum_{i=0}^{N-1} qx_i^2 + ru_i^2 \\ \text{s.t.} \quad & x_{i+1} = 1.5x_i + u_i \\ & -0.1 \leq u_i \leq 0.1 \\ & x_0 = x(k) \\ & x_N \in \mathcal{X}_f, \end{aligned}$$

where $U = [u_0, \dots, u_{N-1}]^T$, $q \geq 0$, $p > 0$, and $r > 0$. What is the largest terminal invariant set $\mathcal{X}_f$ for the problem under a stabilizing linear terminal control law, i.e., $u(k) = Kx(k)$? (For this exercise, Lyapunov stability is sufficient for a controller to be stabilizing, that is, we don't require it to result in an asymptotically stable closed loop.)

1. $[-0.15, 0.15]$
2. $[-1.5, 1.5]$
3. $[-0.2, 0.2]$
4. $[-0.5, 0.5]$

Hint: Algorithm to compute maximum positively invariant set Ω_∞ :

1. init $\Omega_0 \leftarrow \mathcal{X}$
2. $\Omega_{i+1} \leftarrow \text{pre}(\Omega_i) \cap \Omega_i$
3. if $\Omega_{i+1} = \Omega_i$ then return $\Omega_\infty = \Omega_i$, else go to 2

Exercise 3

Consider the double integrator system

$$x(k+1) = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} \frac{T_s^2}{2} \\ T_s \end{bmatrix} u(k)$$

with $T_s = 0.1$ and box constraints $|u| < 0.1$, $\|x\|_\infty \leq 1$ component-wise.

- a) Using the MPT toolbox (<https://www.mpt3.org/>) compute the maximum control invariant set.
- b) Design a linear state feedback controller Kx and compute the maximum invariant set for the closed-loop system, resulting in an inner approximation of the control invariant set.
- c) Investigate these computations for a triple integrator system